Aman Hamidsha

phone | email | linkedin | github | website

Education

University of Warwick Grade: **2:1**
Bachelor of Science in Computer Science *2023-2026*

Merryland Interational School Grades: **4A*1A** (A-levels)
Top in the UAE for AS Level Further Mathematics *2021-2023*

Projects

Awar-E | *Dart (Flutter, Riverpod, Postgres, Serverpod)* Jan - Mar 2026

- Developed a full-stack cross-platform application using Flutter (web, linux, and android compatible).
- Contains simulations to help the general public gain e-safety awareness and to be cautious of cyber threats. (simulators: email, SMS, → threats: phishing, smishing)
- Implemented OAuth for secure logins.
- Implemented backend using postgres and serverpod (account management and logins).
- Dyanmically displays up to date cyber news and CVE infomation baesd on keywords via the CVE API.
- Developed a machine learning algorithm to analyse user responses to suspicious emails and texts.
- App content (curriculum) baesd on NIST publications.

SBC-ARP | *C (sys, net, netinet, netpacket, openssl, arpa)* Jul - Sep 2025

- Researched designing a secure, backwards-compatible ARP protocol to mitigate ARP spoofing and man-in-the-middle attacks while preserving compatibility with existing network infrastructure.
- Developed a decentralised IP–MAC binding verification scheme using cryptographically signed DHCP lease ”tokens” (RSA signatures), enabling distributed validation across network switches and clients.
- Evaluated and compared centralised (DHCP server validation) and decentralised trust models, analysing trade-offs including scalability, bandwidth usage, single points of failure, and resilience against DHCP spoofing.
- Proposed network-layer security mechanisms including Dynamic ARP Inspection, trusted/untrusted switch ports, and authenticated IP–MAC binding verification to strengthen enterprise network security.
- Collaborated with two colleagues as part of Warwick's Undergraduate Research Support Scheme (URSS). Received maximum bursary of £1500.

Technical Skills

Languages: Python, C, Rust, Haskell, OCaml, Dart
Developer Tools: Git, Docker, VS Code